Northeastern University
Electrical and Computer Engineering

EECE 6544: Introduction to Machine Learning and Pattern Recognition

Summer 2026

Assignment #06

Household Energy Consumption Forecasting

The problem: You work as a data scientist for a smart-grid utility company. The company wants to forecast each household's electricity consumption for the next 24 hours so it can balance load on the grid, offer dynamic pricing, and alert customers about unusual usage spikes (which can indicate faulty appliances). Since power consumption is sequential data with daily and seasonal patterns plus long-term dependencies (e.g., weekday vs. weekend habits, winter heating cycles), an LSTM is a natural fit it can remember patterns across long time windows better than classical models like ARIMA.

Your task:

Build an LSTM model that takes the past 7 days of minute-level (or hourly-aggregated) power readings and predicts the next 24 hours of consumption.

Public Dataset

Individual Household Electric Power Consumption (UCI Machine Learning Repository) — https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption

It contains ~2 million measurements from one household in France over 47 months (Dec 2006 – Nov 2010), sampled every minute, with features like global active power, voltage, current intensity, and three sub-metering channels (kitchen, laundry, water heater/AC).

5 Requirements for This Real-World Problem

1. **Data preprocessing and gap handling** the system must preprocess the raw sensor data before training. You should identify and handle the approximately **1.25% missing values** (caused by sensor outages) by selecting and justifying an appropriate imputation technique, such as **interpolation, forward-fill, backward-fill, or another suitable method**. You should also determine an appropriate strategy for **resampling the minute-level readings into hourly aggregates** (mean, median, maximum, minimum, or another justified aggregation method). Their preprocessing choices should be clearly explained and supported by their analysis.

2. **Prediction accuracy** The model must forecast the next 24 hours of hourly consumption with a Mean Absolute Percentage Error (MAPE) of $\leq 10\%$ on the held-out test set (the final 6 months of data, never seen during training).

3. **Temporal integrity** Train/validation/test splits must be chronological (no random shuffling across time), and all feature scaling (MinMax normalization) must be fit only on the training set to prevent data leakage from the future.

4. **Inference latency** For deployment on the utility's servers, a single 24-hour forecast for one household must be generated in under 500 ms, so predictions for thousands of households can be refreshed every hour.

5. **Anomaly alerting** When actual consumption deviates from the forecast by more than a defined threshold (e.g., 3 standard deviations of historical residuals) for 2+ consecutive hours, the system must flag the household for a usage alert, with a false-alarm rate below 5%.

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